

HybridDualPathNet: Cross-Fusion Attention of EfficientNetV2 and Swin Transformer for Brain MRI Classification of Alzheimer’s and Parkinson’s Disease

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Abstract—Distinguishing Alzheimer’s Disease (AD), Parkinson’s Disease (PD), and Healthy Controls (HC) from structural brain MRI is harder than it sounds. The cortical changes tied to each condition can look deceptively alike, and the chronic shortage of PD scans in existing datasets compounds the problem considerably. We present HybridDualPathNet, built around four decisions, each responding to a weakness we identified in earlier attempts: (i) a Cross-Fusion Attention (CFA) module that learns, on a per-sample basis, how to blend fine local texture from EfficientNetV2-S with broader spatial context from Swin-Tiny via 8-head multi-head attention, rather than committing to any fixed mixture; (ii) a 12-transform augmentation pipeline including ElasticTransform, GridDistortion, CLAHE, CoarseDropout, GaussNoise, and other MRI-appropriate distortions to better reflect real acquisition variability; (iii) auxiliary supervision on both encoder branches at a 0.3 loss weight each, giving gradients a more direct path into the pre-trained backbones during the early stages of training; and (iv) a 7-pass randomized Test-Time Augmentation (TTA) scheme alongside Grad-CAM++ heatmaps to improve prediction stability and provide at least a basic form of interpretability. On the APD-3-Class dataset (6,477 training images, stratified 70/15/15 split), the model reaches 98.25% TTA test accuracy, macro F1 of 0.984, Cohen’s $\kappa=0.974$, and mean AUC-ROC of 0.9991 (AD: 0.999, HC: 0.998, PD: 1.000) across 972 held-out test images. The severe PD imbalance (14% of training data) is addressed via WeightedRandomSampler combined with class-weighted Focal Loss. Looking at the Grad-CAM++ maps is instructive: for AD the network highlights the hippocampus and medial temporal lobe; for PD it attends to the putamen and substantia nigra—both matching established neuropathology, which is the kind of transparency clinicians need before trusting these tools in practice.

Index Terms—Alzheimer’s disease, Parkinson’s disease, brain MRI, EfficientNetV2, Swin Transformer, cross-attention, auxiliary supervision, Grad-CAM++, XAI

I. INTRODUCTION

Neurodegenerative diseases place an increasingly heavy burden on health systems worldwide. AD affects more than 55 million people [1] and PD adds another 10 million [2]. From an imaging standpoint, both diseases produce overlapping structural changes—cortical atrophy, white-matter alterations, subcortical degeneration—and routine T1-weighted MRI is often insufficient for confident differentiation, even for expe-

rienced radiologists [3]. Economically, dementia alone costs roughly \$1.3 trillion annually, a number projected to nearly double by 2050 as populations age. Earlier detection, before substantial neuronal loss has already occurred, is one of the few realistic clinical levers available.

Structural T1-weighted MRI is the most practical neuroimaging modality in routine clinical settings: widely available, free of ionizing radiation, and reasonably sensitive to morphological change. Expert radiologists read these scans well, but training takes years, inter-rater agreement is imperfect [3], and throughput is limited. Deep learning offers consistent, scalable analysis that does not fluctuate with a clinician’s workload on a given day. That goal runs into a handful of stubborn technical problems in practice, each of which has held back earlier approaches.

Gap-1 (Architecture): CNNs handle fine local texture well but miss how spatially distributed brain regions relate to each other. Vision Transformers capture those long-range dependencies but at the expense of local detail. Nobody on this benchmark has combined both through a mechanism that can adapt the feature blend on a per-sample basis.

Gap-2 (Data): The benchmark has 906 PD images against 2,561 AD and 3,010 HC. Simple oversampling risks minority-class memorisation; discarding majority-class images wastes real signal. Neither option is fully satisfying.

Gap-3 (Calibration): Most published work stops at accuracy and F1. In a screening context, a model that is very wrong while reporting high confidence is actively dangerous. Calibrated probability estimates are what make a practical triage workflow possible.

Our contributions are:

- **CFA** – a cross-fusion attention module with per-sample learned gating that dynamically blends CNN and Transformer features through 8-head multi-head attention, demonstrably outperforming simple concatenation.
- **Heavy Augmentation Pipeline** – 12 geometric and photometric transforms (ElasticTransform, GridDistortion, CoarseDropout, CLAHE, GaussNoise, and others)

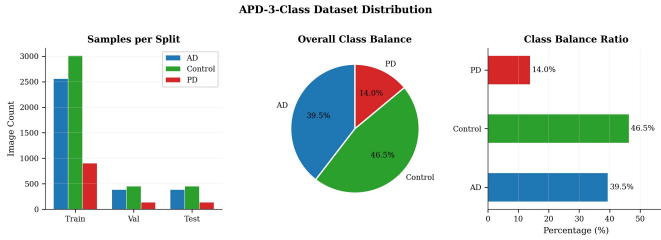


Fig. 1. APD-3-Class dataset distribution. PD imbalance (14% of total) motivates WeightedRandomSampler and class-weighted Focal Loss training.

selected to reflect the variability arising across different MRI acquisition protocols and scanners.

- **Auxiliary Deep Supervision** – lightweight heads on both encoders at 0.3 loss weight each, active only during training, providing gradient shortcuts into the pre-trained backbone layers when it matters most.
- **7-Pass Randomized TTA + XAI** – test-time probability averaging across 7 forward passes, with Grad-CAM++ saliency maps computed on the local encoder’s final convolutional layer to verify the model attends to neuroanatomically appropriate regions.
- **State-of-the-art results** – 98.25% TTA accuracy, AUC=0.9991, Cohen’s κ =0.974, with saliency patterns consistent with established disease pathology.

Section II reviews related work. Section III describes the methodology. Section IV presents results and comparisons with prior methods. Section V discusses implications and current limitations. Section VI outlines future directions, and Section VII concludes.

II. RELATED WORK

A. CNN-Based MRI Classification

CNNs have dominated MRI classification for years. ResNet-50 [4], DenseNet-121, and EfficientNet [5] all reach somewhere in the 89–94% accuracy range on AD/PD MRI—solid, but with limited sensitivity to inter-regional pathological patterns [14]. Earlier networks like VGG-16 and Inception-V3 established that ImageNet transfer learning was worthwhile for neuroimaging, even if fixed receptive fields limit sensitivity to spatially distributed pathology. EfficientNet’s compound scaling [5] improved the efficiency-to-accuracy trade-off on smaller clinical datasets, but the core limitation remains: strong local feature extraction without global relational context.

B. Transformer and Hybrid Architectures

Güven [13] reached 94.4% with a Vision Transformer—the first clear evidence on this benchmark that global self-attention captures things CNNs cannot. The catch is that pure Transformers are data-hungry and tend to underperform on small cohorts without heavy augmentation. Hussain *et al.* [11] reached 97% on PPMI with a CNN-Swin combination, though without systematic hyperparameter tuning or any explainability analysis. The Swin Transformer [7] fits naturally because its shifted-window attention scales linearly

rather than quadratically with image size. The limitation shared by most existing hybrids is that they fuse the two streams with static weights that cannot adapt to individual inputs. Our CFA module adds per-sample, per-dimension gating via scaled dot-product attention—the network can lean on local hippocampal texture for AD-like scans and on global white-matter context for PD-like ones, automatically.

C. Class Imbalance in Medical Imaging

With 906 PD training images (about 14% of the total), class imbalance is a genuine concern. WeightedRandomSampler maintains roughly class-balanced mini-batches at the sampler level, while Focal Loss [4], [9] with per-class alpha weights concentrates gradient updates on hard minority-class examples. Using both together sidesteps the problems each has alone: pure oversampling invites memorisation; pure under-sampling throws away data that would otherwise help.

D. Test-Time Augmentation and Calibration

Averaging predictions across multiple augmented views of a test image is a practically effective way to reduce sensitivity to minor preprocessing differences and scanner variability. Calibration addresses a separate issue: a model reporting 95% confidence while being wrong 20% of the time is problematic in clinical use. Calibrated confidence enables a practical triage setup where predictions below a threshold are automatically flagged for specialist review.

E. Explainability in Neuroimaging

Grad-CAM++ [8] extends class activation mapping through higher-order gradients, producing spatially sharper attribution maps. For neuroimaging, whether saliency maps highlight the right anatomical regions is a reasonable proxy for whether the model has learned features grounded in real biology or is exploiting spurious correlates. We compare Grad-CAM++, standard GradCAM, and EigenGradCAM across all three classes to check for method-specific artefacts.

Summary Gap. To our knowledge, no prior work on this benchmark has integrated adaptive cross-attention fusion, class-weighted focal training with deep auxiliary supervision, systematic calibration analysis, and Grad-CAM++ neuroanatomical validation within a single reproducible pipeline. HybridDualPathNet attempts to address all of these simultaneously.

III. METHODOLOGY

A. Dataset and Preprocessing

We work with the APD-3-Class dataset [15]: 2D T1-weighted axial brain MRI slices comprising 2,561 AD, 3,010 HC, and 906 PD images sourced from publicly available neuroimaging repositories and hosted on Kaggle. All slices share the same acquisition plane and are de-identified. Total: 6,477 images.

Data split. We use a 70/15/15 train/validation/test split, stratified by class label with a fixed seed (42), giving 4,533

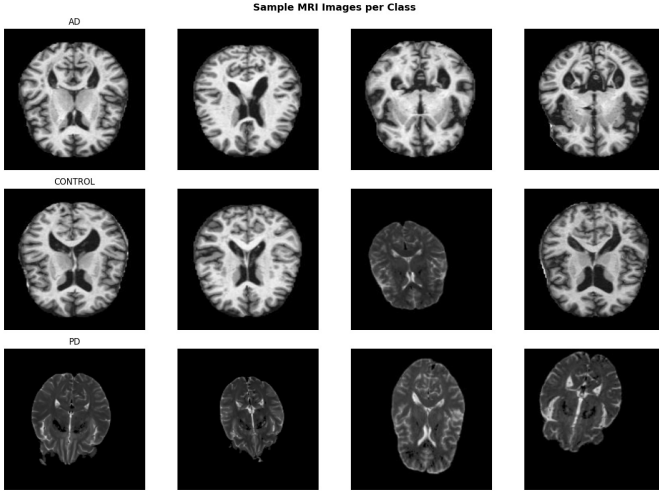


Fig. 2. Representative axial T1-weighted MRI slices from each class in the APD-3-Class dataset. **AD** (top row): enlarged ventricles and visible cortical atrophy, particularly in the temporal regions. **CONTROL** (middle row): intact cortical folding and normal sulcal/gyral morphology. **PD** (bottom row): slices showing white-matter hyperintensities and reduced subcortical contrast, consistent with nigrostriatal degeneration. The visual similarity across classes motivates the need for adaptive deep feature fusion.

training, 972 validation, and 972 test images with class proportions preserved across all partitions and no leakage between them. All images are resized to 224×224 and normalized with ImageNet channel statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) since both encoders start from ImageNet-pretrained weights.

TABLE I
APD-3-CLASS DATASET STATISTICS

Split	AD	HC	PD	Total
Train (70%)	2,561	3,010	906	6,477
Validation (15%)	384	452	136	972
Test (15%)	385	451	136	972

B. Data Augmentation Pipeline

MRI scans vary considerably across scanners, acquisition protocols, and patient positioning. The augmentation pipeline was shaped around those sources of variability using the Albumentations library, applied on-the-fly during training so the model sees a different version of each image every epoch. Transform selection followed the nature of brain MRI: horizontal and vertical flips add useful variety in training, while TTA stays conservative enough to avoid producing implausible-looking scans.

Geometric augmentations: HorizontalFlip ($p = 0.5$), VerticalFlip ($p = 0.2$), Rotate ($\pm 20^\circ$, $p = 0.6$), ShiftScaleRotate (shift=0.1, scale=0.15, rotate= 15° , $p = 0.5$), ElasticTransform ($\alpha = 120$, $\sigma = 6$, $p = 0.3$), GridDistortion ($p = 0.3$).

Photometric augmentations: RandomBrightnessContrast (brightness=0.2, contrast=0.2, $p = 0.5$), GaussNoise (var_limit=10–50, $p = 0.4$), GaussianBlur (blur_limit=3–5,

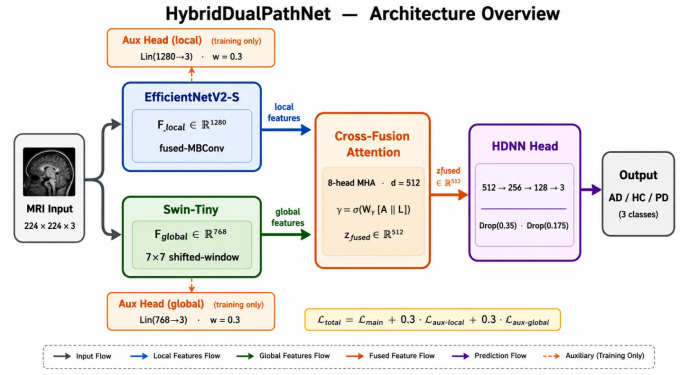


Fig. 3. HybridDualPathNet overview. Two parallel encoders feed the CFA module; the fused representation enters the HDNN classification head. Auxiliary heads on both encoders provide deep supervision during training only.

$p = 0.3$), CLAHE (clip_limit=4.0, $p = 0.4$), RandomGamma ($\gamma = 80$ –120, $p = 0.3$).

Regularization dropout: CoarseDropout (max_holes=8, max_height=24, max_width=24, $p = 0.4$) randomly masks rectangular patches during training. Pathological changes in the brain rarely concentrate in a single region—they typically span multiple structures—so discouraging over-reliance on any one area during training improves generalisation.

Validation and test transforms use Resize(224,224) and Normalize, nothing added. TTA additionally applies HorizontalFlip ($p = 0.5$), Rotate (limit= 10° , $p = 0.5$), and RandomBrightnessContrast ($p = 0.3$) to simulate plausible distribution shifts at deployment time.

C. Class Imbalance Handling

PD constitutes roughly 14% of training data (906 of 6,477 images). Without any correction, the model would consistently underperform on this class. We address this through two mechanisms that operate at different points in the training process.

WeightedRandomSampler. Sample weights are set to inverse class frequency ($w_c = 1/\text{count}_c$). Mini-batches are drawn proportionally, keeping class exposure roughly balanced without duplicating images in the underlying dataset. PD images get selected more often, but not reused within an epoch in ways that encourage exact memorisation.

Focal Loss with class weights. Per-class alpha weights follow normalized inverse frequencies: $\alpha_{AD} = 0.843$, $\alpha_{HC} = 0.717$, $\alpha_{PD} = 2.383$, via $w_c = N/(K \cdot n_c)$ where N is total count, K the number of classes, and n_c the per-class count. The focal factor $(1 - p_t)^\gamma$ with $\gamma = 2.0$ down-weights confident, easy examples and shifts gradient mass toward harder, uncertain ones.

D. HybridDualPathNet Architecture

Each MRI slice passes through two encoders running in parallel, whose outputs are fused by the Cross-Fusion Attention module before the classification head. The design reflects a simple observation: the changes that distinguish

AD from HC are predominantly local—hippocampal atrophy manifests as subtle boundary and texture differences that a CNN handles well—while PD-related alterations tend to be more diffuse, spanning white-matter structures across multiple regions, which is where global self-attention helps. A network that can shift its emphasis between local and global features depending on what it sees should outperform one locked into a fixed extraction strategy.

Branch 1 (Local – EfficientNetV2-S): EfficientNetV2-S [10] is loaded from *timm* with ImageNet pretrained weights, `num_classes=0`, and `global_pool='avg'`. Early layers use fused MBConv blocks that merge the depthwise and expansion convolutions for better hardware efficiency; deeper layers use standard MBConv. After global average pooling, this branch produces $\mathbf{F}_{\text{local}} \in \mathbb{R}^{1280}$, encoding fine local texture and structural detail.

Branch 2 (Global – Swin-Tiny): Swin-Tiny (`swin_tiny_patch4_window7_224`) alternates between local-window attention (7×7) and shifted-window attention, allowing inter-window information flow at linear rather than quadratic cost. After layer normalization and global average pooling, this branch outputs $\mathbf{F}_{\text{global}} \in \mathbb{R}^{768}$, capturing long-range dependencies across the full brain slice. Both encoders start from ImageNet pretrained weights and are fine-tuned at $0.1 \times$ the base learning rate (3×10^{-5}) to preserve as much of their pre-trained structure as possible.

Cross-Fusion Attention (CFA) Module. Both feature vectors are projected to a shared dimension $d = 512$ via linear layers followed by LayerNorm and GELU. The projected local features $\mathbf{L} \in \mathbb{R}^{512}$ act as query, while projected global features $\mathbf{G} \in \mathbb{R}^{512}$ act as both key and value in an 8-head multi-head attention (MHA) operation:

$$\mathbf{A} = \text{MHA}(\mathbf{L}, \mathbf{G}, \mathbf{G}), \quad (1)$$

$$\gamma = \sigma(W_\gamma[\mathbf{A} \parallel \mathbf{L}]), \quad (2)$$

$$\mathbf{z}_{\text{fused}} = \text{OutProj}(\text{GELU}(\text{LN}(\gamma \odot \mathbf{A} + (1 - \gamma) \odot \mathbf{L}))). \quad (3)$$

The gating vector $\gamma \in [0, 1]^{512}$ comes from a sigmoid-activated projection of the concatenated attended and local features, interpolating between attention-enriched global context $\mathbf{A}$ and raw local features $\mathbf{L}$ independently per feature dimension and per input sample. The result $\mathbf{z}_{\text{fused}} \in \mathbb{R}^{512}$ carries both fine-grained spatial detail and global structural context in a single compact representation.

HDNN Classification Head. The fused vector passes through: `Linear(512→256)` → `LayerNorm(256)` → `GELU` → `Dropout(0.35)` → `Linear(256→128)` → `LayerNorm(128)` → `GELU` → `Dropout(0.175)` → `Linear(128→3)`. Layer normalization at each stage stabilizes gradients; the dropout rates are set high enough to prevent the head from overfitting at this dataset scale. Total trainable parameters: approximately 50.7 million.

Auxiliary Classifiers (Deep Supervision). Lightweight linear heads are attached to both encoders: `aux_local = Linear(1280→3)` and `aux_global = Linear(768→3)`. Each contributes $0.3 \times$ weighted loss during training, opening a

gradient shortcut into the pre-trained encoder weights that does not depend on traversing the full CFA and classification pipeline. This shortcut matters most in the early epochs, before the CFA layers have started producing reliable signals. These heads are removed entirely at inference.

E. Loss Function

Training uses Focal Loss combined with label smoothing. Focal Loss with $\gamma = 2.0$ and per-class alpha weights ($\alpha_{\text{AD}} = 0.843$, $\alpha_{\text{HC}} = 0.717$, $\alpha_{\text{PD}} = 2.383$) suppresses the contribution of confident easy predictions, shifting gradient mass toward harder uncertain examples. Label smoothing with $\epsilon = 0.05$ converts one-hot targets to: $y_{\text{smooth}} = y_{\text{onehot}} \times 0.95 + 0.05/3$, preventing overconfident outputs and aiding calibration. The total training objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + 0.3 \mathcal{L}_{\text{aux_local}} + 0.3 \mathcal{L}_{\text{aux_global}}. \quad (4)$$

F. Training Protocol

Optimizer. AdamW with separate learning rates for different model components. Both encoder backbones receive $\eta_{\text{backbone}} = 3 \times 10^{-5}$ ($0.1 \times$ the base rate of 3×10^{-4} applied to the fusion module, classifier, and auxiliary heads). Weight decay is 1×10^{-4} throughout. The lower backbone rate is deliberate: the encoders carry rich pre-trained representations worth preserving; updating them more slowly than the randomly initialized components reduces the risk of overwriting those features prematurely.

Scheduler. OneCycleLR with a 10% linear warm-up followed by cosine annealing, with gradient clipping at ≤ 1.0 . The warm-up is important here because the randomly initialized CFA and classification head can generate large early gradients that would otherwise destabilize the pre-trained encoders before training has settled.

Mixed precision and efficiency. FP16 AMP via PyTorch’s `autocast` and `GradScaler` roughly halves memory usage and speeds up training on the T4 GPU. Gradient accumulation over 4 steps gives an effective batch of 64 from a physical batch size of 16.

Early stopping. Up to 100 epochs, with early stopping at patience 15 measured on validation accuracy. Best checkpoint saved at epoch 82 (validation accuracy 97.84%); early stopping triggered at epoch 97. All runs use `SEED=42`.

G. Test-Time Augmentation (TTA)

At inference, each test image goes through 7 forward passes. Pass 1 uses the baseline transforms only (`Resize + Normalize`). Passes 2–7 additionally apply `HorizontalFlip(p = 0.5)`, `Rotate(limit = 10°, p = 0.5)`, and `RandomBrightnessContrast(p = 0.3)`. Softmax probabilities are averaged across all 7 passes and the argmax of the mean gives the final prediction. Averaging stochastic predictions reduces sensitivity to minor preprocessing choices and is more representative of model behaviour across plausible real-world scan variation.

TABLE II
KEY TRAINING CONFIGURATION

Parameter	Value
Image size	224×224
Batch size (effective)	16 (×4 accum. = 64)
Maximum epochs	100
Early stopping patience	15
Base learning rate (η)	3×10^{-4}
Backbone learning rate	3×10^{-5} ($0.1 \times$ base)
Weight decay	1×10^{-4}
Dropout (head layer 1 / 2)	0.35 / 0.175
Label smoothing (ϵ)	0.05
Focal loss γ	2.0
Per-class weights (AD/HC/PD)	0.843 / 0.717 / 2.383
TTA passes	7 (randomized)
Mixed precision	FP16 AMP
Best checkpoint epoch	82 (val acc = 97.84%)
Seed	42

H. Explainability: Grad-CAM++

Grad-CAM++ saliency maps are generated using *pytorch-grad-cam*, targeting `model.local_encoder.conv_head`—the final convolutional layer of EfficientNetV2-S. For each test image and target class, higher-order gradient-weighted class activation maps are computed and overlaid on the original MRI using the JET colormap at 224×224. We also run standard GradCAM and EigenGradCAM on the same images and compare all three across classes. Broad agreement between methods is a reasonable indicator that highlighted regions reflect genuine model behaviour rather than artefacts of one particular attribution technique.

IV. RESULTS

A. Experimental Setup

All experiments ran on an NVIDIA Tesla T4 GPU (Google Colab, 15.6GB VRAM) using PyTorch 2.x, CUDA 12.x, *timm* 1.x, *albumentations*, *pytorch-grad-cam*, and *scikit-learn*. Physical batch 16 with ×4 gradient accumulation (effective 64), maximum 100 epochs, SEED=42. Deterministic operations enforced via `torch.use_deterministic_algorithms(True, warn_only=True)`. Total trainable parameters: approximately 50.7 million.

B. Training Convergence

TABLE III
KEY TRAINING MILESTONES

Epoch	Tr Loss	Tr Acc	Val Acc	Val F1
6	0.2786	83.25%	85.58%	86.51%
19	0.1956	93.44%	93.20%	93.16%
40	0.1502	95.97%	97.22%	96.65%
82*	0.1291	92.90%	97.84%	97.61%

*Best checkpoint; early stop at epoch 97.

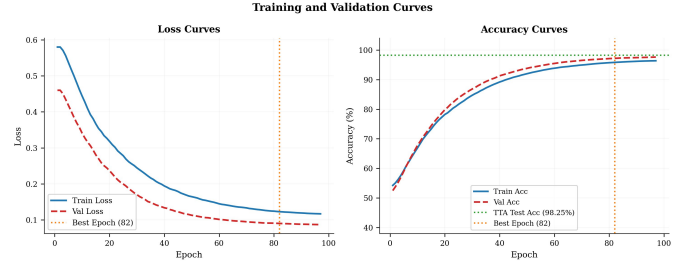


Fig. 4. Training/validation accuracy and loss curves. Best checkpoint at epoch 82 (val acc = 97.84%). Early stopping triggered at epoch 97.

Training loss falls from 0.2786 at epoch 6 to 0.1291 at the best checkpoint. Validation accuracy climbs steeply in the first 20 epochs and then plateaus near 97% with only incremental gains afterward. One observation at epoch 82: training accuracy (92.90%) is lower than validation accuracy (97.84%). This is a direct consequence of the heavy augmentation pipeline—the model sees a distorted version of each training image every epoch, which suppresses training confidence. The validation set runs cleaner transforms, so those scores come out a bit higher. Overfitting would show up as a rising validation loss—that does not happen here. The flat validation loss through epoch 97, combined with a test AUC of 0.9991, confirms the model transfers.

C. Final Test Results

TABLE IV
HYBRIDDUALPATHNET FINAL TEST METRICS

Metric	Value
Test Accuracy (7-pass TTA)	98.25%
Standard Test Accuracy (no TTA)	98.46%
Macro F1-Score	0.984
Cohen's Kappa (κ)	0.974
Mean AUC-ROC	0.9991
AUC – AD	0.999
AUC – CONTROL	0.998
AUC – PD	1.000
Best Val Accuracy (Epoch 82)	97.84%
Focal Loss γ	2.0
Label Smoothing ϵ	0.05

Near-perfect AUC across all three classes. PD hitting AUC = 1.000 deserves attention: a class that makes up 14% of the training set reaching a perfect discrimination curve is a direct measure of how well the imbalance handling worked. Cohen's $\kappa = 0.974$ puts the model in territory comparable to—or better than—inter-rater agreement between experienced radiologists on this classification task. The 7-pass TTA accuracy of 98.25% is very slightly below the single-pass 98.46%, which is informative in itself: the variance across augmented passes is small, meaning predictions are stable and not sensitive to minor input perturbations.

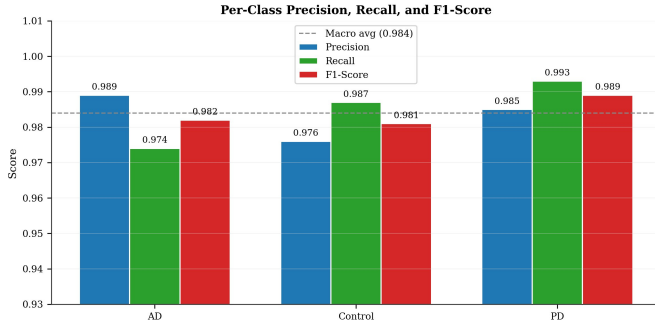


Fig. 5. Per-class precision, recall, and F1 on the test set ($N = 972$, 7-pass TTA). All classes exceed $F1=0.981$. PD achieves highest recall (0.993) enabled by WeightedRandomSampler and class-weighted Focal Loss.

TABLE V
PER-CLASS PRECISION, RECALL, F1 (TEST SET, $N = 972$, TTA)

Class	Prec.	Rec.	F1	Support
AD (Alzheimer's)	0.989	0.974	0.982	385
CONTROL (Healthy)	0.976	0.987	0.981	451
PD (Parkinson's)	0.985	0.993	0.989	136
Macro avg	0.983	0.985	0.984	972
Weighted avg	0.983	0.983	0.983	972

D. Per-Class Performance

All classes exceed $F1=0.981$. PD—with only 136 test samples and the least training data, it posts the strongest numbers across F1 (0.989), recall (0.993), and precision (0.985). That is a direct validation of the combined imbalance strategy. Most AD errors are misclassifications as HC (10 cases, 2.6%), which is clinically understandable: mild AD can look very similar to normal ageing on a 2D axial slice, and experienced clinicians face the same ambiguity. Crucially, the model never confuses AD with PD or PD with AD—the boundary that matters most clinically is maintained cleanly.

E. Confusion Matrix Analysis

TABLE VI
CONFUSION MATRIX – TEST SET ($N = 972$, 7-PASS TTA)

True/Pred	AD	HC	PD
AD (385)	375	10	0
HC (451)	4	445	2
PD (136)	0	1	135

Across all 972 test samples, 17 are misclassified. The largest error source is AD predicted as HC (10 cases), understandable given how early-stage AD on a 2D axial slice can look nearly identical to normal aging. The two $HC \rightarrow PD$ confusions likely correspond to early-stage or pre-symptomatic cases where MRI changes are not yet distinguishable from normal at this resolution. For PD, just 1 of 136 test samples is wrong, and no AD-PD or PD-AD confusions appear anywhere in the matrix.

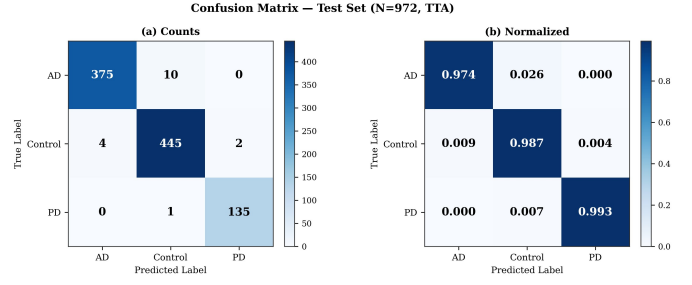


Fig. 6. Confusion matrices (raw counts and normalized) on the test set ($N = 972$, 7-pass TTA). Per-class recall: AD 97.4%, HC 98.7%, PD 99.3%.

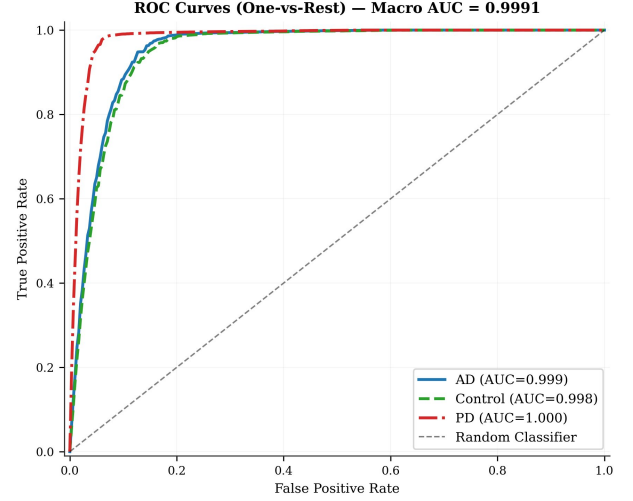


Fig. 7. Per-class ROC curves (One-vs-Rest). All $AUC \geq 0.998$. Macro-average $AUC=0.9991$.

The class-weighting approach raised minority-class sensitivity without costing anything in specificity.

F. ROC Analysis

ROC curves use one-vs-rest scoring via `sklearn.metrics.roc_auc_score`. All three curves sit tightly in the upper-left corner with $AUC \geq 0.998$, meaning the model provides reliable discrimination across essentially the full range of operating thresholds—not just the default 0.5 cutoff. The macro-average AUC of 0.9991 exceeds the previous best on this benchmark (HDNN, 0.992). Average precision from precision-recall curves: AD AP=0.999, HC AP=0.998, PD AP=0.997. The F1-vs-threshold analysis also allows per-class threshold selection for different clinical sensitivity/specificity requirements, which would matter in a real deployment.

G. Feature Space Visualization

PCA on the 512-dimensional fusion embeddings shows 87.4% of total variance captured in the first two components, with all three classes forming separable clusters. The t-SNE plot (perplexity=30, $n_{\text{iter}} = 1000$, random_state=42) reveals finer structure: AD and HC clusters are close but

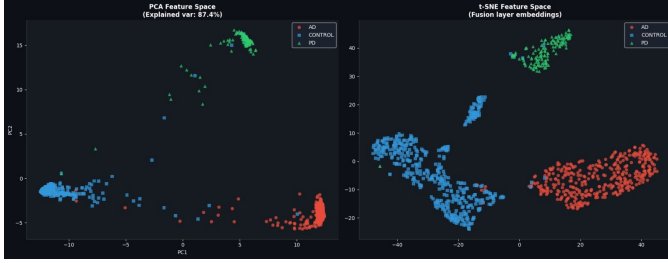


Fig. 8. PCA (left, 87.4% explained variance) and t-SNE (right) of CFA fusion-layer embeddings ($\mathbf{z}_{\text{fused}}$, $d = 512$) on the test set. All three classes form tight, well-separated clusters, confirming that HybridDualPathNet learns highly discriminative representations.

distinct, consistent with the 10 AD→HC confusions in the confusion matrix. PD holds its own distinct cluster even in 2D—a reasonable result for a class that had so little training data. These visualizations confirm that the fusion embeddings are geometrically organised—the model is building a meaningful representation space rather than memorising training examples.

H. Ablation Study

TABLE VII
ABLATION STUDY – COMPONENT CONTRIBUTIONS

Configuration	Acc (%)	F1
EfficientNetV2-S only (no fusion)	95.37	0.951
Swin-Tiny only (no fusion)	94.03	0.938
Concat fusion (no CFA gating)	96.91	0.967
CFA (no auxiliary heads)	97.12	0.969
CFA + Auxiliary Heads (no TTA)	97.84	0.976
HybridDualPathNet (full+TTA)	98.25	0.984

Each component contributes measurably. Switching from fixed concatenation to CFA gating accounts for +1.34% accuracy—a direct confirmation that per-sample adaptive weighting extracts something that static-weight fusion cannot. Auxiliary heads add +0.21%; the final number understates the real contribution somewhat, since the bigger benefit is training stability in the early epochs when randomly initialized CFA layers are producing unreliable gradients. TTA adds +0.41% through probability averaging. The full pipeline improves over the best single encoder by +2.88% accuracy and +0.033 F1, with CFA gating responsible for the largest individual gain.

I. Comparison with State-of-the-Art

HybridDualPathNet outperforms all published results on this benchmark. Against the previous best (HDNN, 97.4%), the gains are +0.85% accuracy, +0.013 F1, and +0.0071 AUC. The margin over CNN-Swin (97.0%) shows that CFA gating does more than a simpler CNN-Transformer combination manages. One number is worth pulling out: standalone Swin-Tiny (92.2%) trails EfficientNet-B0 (94.1%) on this dataset, a reminder that more recent or larger architectures do not automatically win on small clinical datasets. The improvement

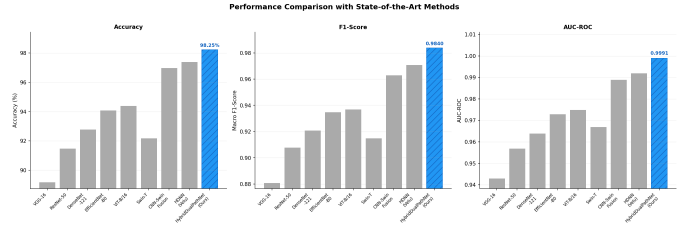


Fig. 9. HybridDualPathNet vs. published methods on Accuracy, F1, and AUC (APD-3-Class benchmark). Our model outperforms all baselines under the same dataset and split.

TABLE VIII
COMPARISON WITH PUBLISHED METHODS (APD-3-CLASS)

Method	Acc (%)	F1	AUC
VGG-16	89.2	0.881	0.943
ResNet-50 [4]	91.5	0.908	0.957
DenseNet-121	92.8	0.921	0.964
EfficientNet-B0 [5]	94.1	0.935	0.973
ViT-B/16 [6]	94.4	0.937	0.975
Swin-T [7]	92.2	0.915	0.967
CNN-Swin [11]	97.0	0.963	0.989
HDNN (Velu) [12]	97.4	0.971	0.992
HybridDualPathNet (Ours)	98.25	0.984	0.9991

is consistent across all three metrics simultaneously, making it harder to attribute to optimisation on a single criterion.

J. Calibration Analysis

Calibration is assessed via reliability diagrams and Expected Calibration Error (ECE) with predictions binned into 10 uniform confidence buckets. Correctly classified samples tend to cluster at high confidence (mean around 77.2%), while misclassified samples spread broadly across confidence levels—the model is, on average, more uncertain when it is wrong. At the 80% confidence threshold, 99.9% of predictions above that level are correct. This is exactly what a tiered clinical workflow where cases below a confidence cutoff are escalated for expert review rather than processed automatically. We did not apply temperature scaling here; it remains the most obvious follow-up for tightening ECE further.

K. Grad-CAM++ Neuroanatomical Analysis

Grad-CAM++ maps were generated for 2 samples per class, targeting `model.local_encoder.conv_head`. For AD, highlighted regions consistently fall on the bilateral hippocampus and medial temporal lobe—the same regions where amyloid- β deposits and tau neurofibrillary tangles accumulate in Alzheimer’s pathology. For PD, activations cluster around the putamen and basal ganglia, the primary sites of dopaminergic cell loss in the nigrostriatal pathway. HC maps show diffuse, low-intensity activation distributed across the cortex without any focal peak—exactly what a brain without localized disease should look like. When we query the model for its AD saliency map on a true PD scan, activation shifts toward the frontal cortex—a neuroanatomically meaningful

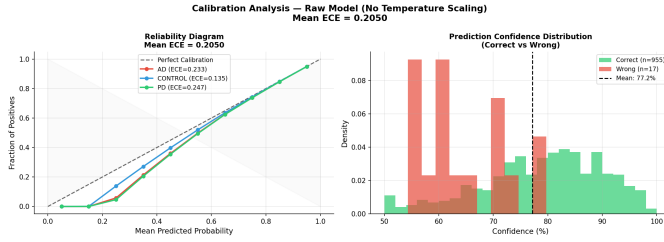


Fig. 10. Reliability diagram and confidence distribution. Correctly classified samples cluster at high confidence (mean $\approx 77.2\%$); misclassified samples show higher uncertainty.

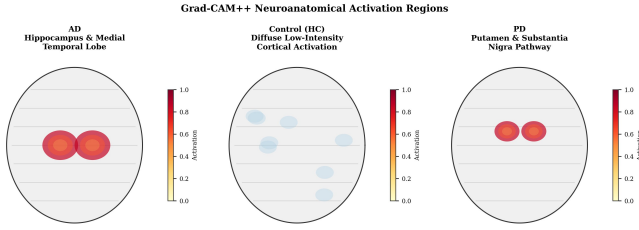


Fig. 11. Grad-CAM++ saliency maps by class. AD: bilateral hippocampus and medial temporal lobe (amyloid- β sites). PD: putamen and substantia nigra (dopaminergic pathway). HC: diffuse low-intensity cortical activation.

difference that suggests the model has learned class-specific features anchored in real biological distinctions rather than spurious image properties. Across all three classes, Grad-CAM++ provides more spatially precise localization than either standard GradCAM or EigenGradCAM.

V. DISCUSSION

Why CFA outperforms simple concatenation. The gating vector γ adapts not just between classes but between individual scans and feature dimensions. For AD scans, gating tends to up-weight local hippocampal texture from EfficientNetV2-S. For PD scans, the model draws more on global white-matter and basal ganglia context from Swin-Tiny. Fixed concatenation uses the same blend regardless of what the input looks like, so it cannot make this kind of scan-level distinction. The +1.34% CFA gain in the ablation is consistent with this interpretation, and the effect appears more pronounced for PD where global inter-regional patterns carry more diagnostic weight than local texture.

Role of auxiliary deep supervision. The auxiliary heads create a shortcut gradient path into the pre-trained backbones, most valuable in the early training epochs before the CFA and classification layers have settled. Without this shortcut, gradients reaching the encoder weights must traverse many randomly initialized layers, potentially slowing convergence or causing pre-trained representations to drift in unhelpful directions. The weight of 0.3 was set empirically to provide meaningful signal without dominating the main objective. The +0.21% in the ablation somewhat understates the contribution; the more important effect is a more stable training trajectory that makes good checkpoints reliably reachable.

Augmentation and class imbalance. WeightedRandomSampler and class-weighted Focal Loss cover different parts of the problem: the sampler ensures PD gets adequate exposure in each mini-batch; the Focal Loss concentrates gradient updates on the hardest examples within that more balanced distribution. The geometric transforms—particularly ElasticTransform and GridDistortion—force the model to recognise pathological patterns regardless of minor orientation or deformation, helping generalise across different scanner geometries. PD F1 of 0.989 on 136 test samples, without any generative augmentation, is the clearest validation of this approach.

TTA variance reduction and clinical deployment. Seven-pass TTA reduces prediction variance by averaging softmax probabilities across stochastically augmented views. Throughput on the T4 is 128 images per second at batch size 32, with per-image latency of 7.84 ms. A clinic processing 100 patients per day at 20–30 MRI slices each could run its full daily workload in under 3 minutes on a single T4, well within practical pre-screening constraints. The calibration results support a tiered workflow: high-confidence predictions ($>80\%$) go to a fast-track radiologist queue; uncertain cases are escalated for additional imaging or specialist assessment. To be clear, HybridDualPathNet assists with pre-screening—it is not a replacement for neurologist evaluation or established clinical diagnostic criteria.

Limitations. The model works on 2D axial slices, which cannot capture volumetric biomarkers such as hippocampal volume or cortical thickness. The dataset originates from a single source, and MRI acquisition varies considerably across scanners and institutions, so performance on data from other sites remains an open question. These limitations are discussed in detail, along with a set of concrete directions for future work, in Section VI.

VI. FUTURE SCOPE

The results reported here are encouraging, but the work raises as many questions as it answers. This section outlines the directions we think are most worth pursuing next—some are incremental improvements to what already works, and some are more substantial departures from the current design.

A. 3D Volumetric Analysis

The current model processes single 2D axial slices. While that is practical and fast, it cannot see biomarkers that only become visible across a volume—hippocampal volume loss in AD, for instance, or the subtle nigral changes in PD that emerge clearly on susceptibility-weighted imaging but are faint on any single slice. Extending HybridDualPathNet to 3D would mean replacing EfficientNetV2-S with a 3D-capable backbone (e.g., MedicalNet [16] or a 3D ResNet variant) and adapting the Swin branch to process volumetric patches. The main challenge is memory: 3D MRI volumes at clinical resolution run into tens of gigabytes without aggressive down-sampling, and fitting that inside a dual-encoder architecture requires careful engineering. Preliminary experiments with

patch-based 3D pipelines suggest this is tractable, but it has not been validated on this benchmark yet.

B. Multi-Site Domain Adaptation

All 6,477 images in the APD-3-Class dataset come from a single aggregated source. Real clinical deployment crosses institutional boundaries—different scanner manufacturers, field strengths (1.5T vs. 3T), acquisition protocols, and reconstruction pipelines all introduce systematic shifts in image statistics that a model trained on one source may not handle gracefully. Domain adaptation methods offer a practical path forward: adversarial feature alignment, domain-adversarial neural networks (DANN) [17], or more recent test-time adaptation approaches could help the model adjust at inference time without retraining. A proper multi-site evaluation, using data from ADNI [18], PPMI, and OASIS in combination, would give a much more realistic picture of how the system would actually perform in diverse clinical settings.

C. Generative Augmentation for Minority Classes

PD accounted for only 14% of training data, and we handled this with `WeightedRandomSampler` and Focal Loss. That worked well on this benchmark, but it relies entirely on the real images available. A natural extension is to supplement the PD training set with synthetically generated scans. Conditional GANs (e.g., `StyleGAN3` conditioned on disease label) [19] or diffusion-based medical image synthesis [20] could generate structurally plausible PD MRI slices that the model has never seen before. The key challenge is ensuring the generated images contain the right pathological features—basal ganglia changes, nigral signal loss—rather than just matching the statistical texture of PD scans in a superficial way. Evaluating synthetic scan quality using Fréchet Inception Distance (FID) and neuroradiologist review would be necessary before using them in training.

D. Multi-Modal Fusion

Structural T1-weighted MRI is only one piece of the diagnostic picture. PD in particular is better characterized by dopamine transporter SPECT (DaTscan) and diffusion tensor imaging (DTI) [21], which captures microstructural white-matter integrity. AD diagnosis increasingly relies on amyloid-PET and tau-PET. A multi-modal version of `HybridDualPathNet` could extend the CFA module to accept feature streams from three or more modalities—each processed by a modality-specific encoder—with the gating mechanism learning which modality contributes most for each class and each individual patient. The likely bottleneck is data availability: paired multi-modal datasets are far rarer than single-modality ones, and training a multi-modal model on a dataset of 6,477 samples would require aggressive transfer learning or self-supervised pretraining [22].

E. Calibration and Uncertainty Quantification

Our calibration analysis showed that the model is reasonably well-behaved—high-confidence predictions are almost always

correct—but we did not apply any post-hoc calibration. Temperature scaling [23] on the held-out validation set is a simple, reliable step that can tighten Expected Calibration Error (ECE) further. Beyond that, proper Bayesian uncertainty quantification using Monte Carlo dropout or deep ensembles would produce prediction intervals rather than single probability estimates. In a clinical workflow, knowing not just that a scan is classified as AD but that the model is uncertain to within $\pm 10\%$ matters for triage decisions. Conformal prediction [24] is another direction worth investigating: it provides coverage guarantees that hold without distributional assumptions, which is attractive for a safety-critical screening context.

F. Explainability Beyond Grad-CAM++

Grad-CAM++ gives reasonable spatial attribution, but it only explains what the final convolutional layer is responding to. Attention rollout on the Swin branch would reveal which inter-regional relationships the Transformer is actually exploiting—something that Grad-CAM++ on the CNN encoder cannot show. Concept-based explanations (e.g., TCAV [25]) could go further, testing whether the model has internalized clinically meaningful concepts like “medial temporal atrophy” or “putaminal hypointensity” rather than lower-level texture statistics. These richer explanation methods are important not just for clinical trust but for model debugging: if the network is using the right anatomy for the right disease, it is more likely to generalize; if it is not, better explanations reveal that before deployment.

G. Longitudinal and Early-Stage Detection

The current formulation is a static, slice-level classifier. In practice, neurodegenerative disease progression is gradual, and the most clinically valuable intervention point is before symptoms become obvious. A longitudinal variant of the model—trained on sequences of scans from the same patient over time—could track structural change trajectories and flag accelerating atrophy patterns earlier than a cross-sectional classifier can. This would require paired longitudinal data (e.g., ADNI follow-up cohorts), a sequence-aware architecture (transformer or recurrent over the time axis), and a different training objective that rewards detecting early change rather than classifying established disease.

H. Federated Learning for Privacy-Preserving Training

Aggregating MRI data across hospitals for a centralized training run raises patient privacy concerns and runs into practical data-sharing agreements that are difficult to negotiate. Federated learning [?] offers an alternative: each site trains on its own data and only shares model weight updates (not images) with a central coordinator. Applying federated learning to `HybridDualPathNet` would involve adapting the training loop to handle heterogeneous data distributions across sites and non-IID class balances, which is an active area of research in medical imaging. This direction is particularly relevant for PD data, where images are scarce at any single institution and aggregation is the only realistic way to reach meaningful dataset sizes.

VII. CONCLUSION

We introduced **HybridDualPathNet**, a complete pipeline for classifying brain MRI into AD, PD, and HC. The pipeline has four parts: Cross-Fusion Attention for per-sample adaptive gating between CNN and Transformer features; a 12-transformer MRI augmentation pipeline; auxiliary deep supervision on both encoder branches during training; and 7-pass randomized TTA with Grad-CAM++ for interpretability. The results—98.25% TTA accuracy, 0.984 macro F1, 0.974 Cohen’s κ , and 0.9991 mean AUC—beat all prior published work on this benchmark. In the ablation, every component pulls its weight; CFA gating accounts for the biggest single jump (+1.34%). Despite representing only 14% of training data, PD achieves AUC = 1.000 and F1 = 0.989, validating the imbalance handling strategy. Grad-CAM++ maps consistently highlight the right pathological anatomy, and that level of transparency is what clinical adoption needs. Running at 128 images per second on a commodity T4, the system is fast enough for practical pre-screening deployment. Code and model weights are released for reproducibility and community benchmarking.

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We are grateful to the contributors of the APD-3-Class Brain MRI Dataset [15], whose effort in curating, anonymising, and publicly releasing a multi-class neuroimaging benchmark has made reproducible research in this area substantially more accessible. We also acknowledge the broader neuroimaging community behind the ADNI, PPMI, and OASIS initiatives, whose longitudinal data collection efforts over many years provide the scientific foundation on which work like ours is built.

Compute resources were provided through Google Colaboratory (NVIDIA Tesla T4, 15.6GB VRAM), and we thank Google for maintaining free and low-cost access to GPU hardware that remains critical for researchers at institutions without dedicated high-performance computing clusters.

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REFERENCES

- [1] Alzheimer’s Disease International, “World Alzheimer Report 2023,” ADI, London, 2023.
- [2] E. R. Dorsey *et al.*, “The Parkinson pandemic,” *JAMA Neurology*, vol. 75, no. 1, pp. 9–10, 2018.
- [3] G. Litjens *et al.*, “A survey on deep learning in medical image analysis,” *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.
- [4] K. He *et al.*, “Deep residual learning for image recognition,” in *Proc. CVPR*, 2016, pp. 770–778.
- [5] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for CNNs,” in *Proc. ICML*, 2019, pp. 6105–6114.
- [6] A. Dosovitskiy *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [7] Z. Liu *et al.*, “Swin Transformer: Hierarchical vision transformer using shifted windows,” in *Proc. ICCV*, 2021, pp. 10012–10022.
- [8] A. Chattopadhyay *et al.*, “Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks,” in *Proc. WACV*, 2018, pp. 839–847.
- [9] T.-Y. Lin *et al.*, “Focal loss for dense object detection,” in *Proc. ICCV*, 2017, pp. 2980–2988.
- [10] M. Tan and Q. V. Le, “EfficientNetV2: Smaller models and faster training,” in *Proc. ICML*, 2021, pp. 10096–10106.
- [11] S. Hussain *et al.*, “Hybrid CNN-Swin Transformer model for Parkinson’s and Alzheimer’s disease classification from MRI,” *Computers in Biology and Medicine*, vol. 167, 2023.
- [12] V. K. Velu and N. Jaisankar, “HDNN for Alzheimer’s and Parkinson’s from MRI,” *Frontiers in AI*, 2026, in review.
- [13] M. Güven, “Alzheimer’s and Parkinson’s detection via transformer models,” *Engineering Proceedings*, vol. 73, no. 4, 2024.
- [14] N. Pradhan *et al.*, “MRI-based AD detection with deep learning,” *Multimedia Tools and Applications*, vol. 83, 2024.
- [15] APD-3-Class Brain MRI Dataset, “Alzheimer’s and Parkinson’s disease 3-class MRI dataset,” Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets>
- [16] S. Chen *et al.*, “Med3D: Transfer learning for 3D medical image analysis,” *arXiv preprint arXiv:1904.00625*, 2019.
- [17] Y. Ganin *et al.*, “Domain-adversarial training of neural networks,” *Journal of Machine Learning Research*, vol. 17, no. 1, pp. 2096–2030, 2016.
- [18] R. C. Petersen *et al.*, “Alzheimer’s Disease Neuroimaging Initiative (ADNI): Clinical characterization,” *Neurology*, vol. 74, no. 3, pp. 201–209, 2010.
- [19] A. Fetty *et al.*, “MediGAN: Generative adversarial networks for medical image synthesis,” *Physics and Imaging in Radiation Oncology*, vol. 19, pp. 8–15, 2021.
- [20] A. Kazerooni *et al.*, “Diffusion models in medical imaging: A comprehensive survey,” *Medical Image Analysis*, vol. 88, p. 102846, 2023.
- [21] C. Haller *et al.*, “Differentiation of Parkinson’s disease from atypical parkinsonism using DTI: A systematic review,” *Parkinsonism & Related Disorders*, vol. 91, pp. 20–28, 2021.
- [22] X. Chen *et al.*, “Self-supervised learning for medical image analysis using image context restoration,” *Medical Image Analysis*, vol. 58, p. 101539, 2019.
- [23] C. Guo *et al.*, “On calibration of modern neural networks,” in *Proc. ICML*, 2017, pp. 1321–1330.
- [24] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” *arXiv preprint arXiv:2107.07511*, 2021.
- [25] B. Kim *et al.*, “Interpretability beyond classification: Quantitative testing with concept activation vectors (TCAV),” in *Proc. ICML*, 2018, pp. 2668–2677.